The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS
NHS Foundation Trust

ANEURYSMAL SUBARACHNOID HAEMORRHAGE (*a*SAH)

Protocol based on agreed guidelines, which may be adapted in individual clinical circumstances with judgement / knowledge of responsible senior clinician

DEFINITION

Subarachnoid haemorrrhage (SAH) is defined as bleeding into the subarachnoid space - therefore into the CSF. Cerebral arteries lie in subarachnoid space and give off small perforating branches to brain tissue. Bleeding from these vessels or from an associated aneurysm occurs primarily into this space. Some intracranial aneurysms are embedded within the brain tissue and their rupture causes intracerebral bleeding with or without SAH. Occasionally, the arachnoid layer is disrupted and a subdural haematoma results. Following SAH, blood within the CSF can disrupt the reabsorption of CSF causing secondary hydrocephalus.

The leading cause of SAH is rupture of an intracranial saccular aneurysm (*a*SAH), a dilatation in the wall of an artery, usually at the junction or bifurcation of arteries. Other causes include anticoagulants, hypertension, arteriovenous malformation (AVM), bleeding diathesis, neoplasm and idiopathic.

CURRENT SAH GUIDELINES

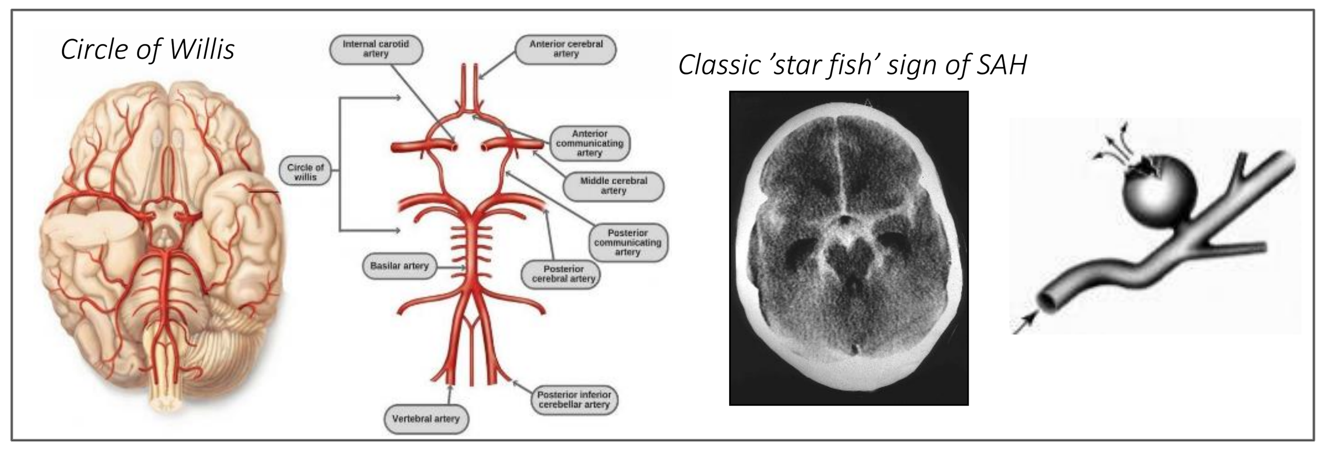
https://www.ahajournals.org/doi/10.1161/STR.0000000000000436#

IMAGING

1. Non-contrast brain CT – *if negative for aSAH, lumbar puncture (LP) to diagnose / exclude SAH*
2. If negative or inconclusive, CT angiography (CTA), digital subtraction angiography (DSA) is indicated to diagnose / exclude cerebral aneurysm
3. DSA is considered the gold-standard investigation and can aid decision-making on optimal treatment

MANAGEMENT TO PREVENT REBLEEDING

- BP should be controlled using short acting medication to prevent severe hypotension, hypertension, and BP variability – aim BP <160mmHg
- Anticoagulant therapy stopped immediately and reversed
- Surgical / endovascular treatment of aneurysm as early as possible following presentation, preferably within 24h of onset, to improve outcome – *to reduce risk of rebleeding and facilitate management of delayed cerebral ischaemia (DCI) (see protocol)*



MANAGEMENT on ADMISSION to NCCU

CLINICAL ASSESSMENT

Neurology

GCS, Pupils, Limbs

- 1 hourly for 4h
- 2 hourly thereafter if stable
- 1 hourly pupil assessment if patient sedated

(Additional monitoring may be needed when neuro exam not possible, as in high grade aSAH)

Vital signs

Heart Rate

- Continuous recording of HR / rhythm
- 12 Lead ECG on admission and daily while QT prolonged
- Avoid drugs that prolong QT interval

Blood Pressure – *Lowering of BP to reduce risk of rebleeding versus risk of secondary ischaemia*

- Avoid hypotension / hypertension / BP variability if possible unsecured aneurysm – *avoid hypertensive peaks*
- Cautious treatment of hypertension – Aim SBP ≤160mmHg *(range: 120-160mmHg)*
- Hypotension *(SBP <120mmHg) – initially treat with IV crystalloid*
- Sustained hypertension *(SBP >160mmHg) – reduction in BP may be achieved with analgesics and nimodipine, but if not, consider short-acting medication (IV infusion of labetalol)*

Respiratory

- Continuous monitoring of SpO_2
- Oxygen therapy if SpO_2 <95%
- If mechanical ventilation for >24h, standardised ICU care bundle recommended to reduce duration of mechanical ventilation / HAP - *lung-protection / low tidal volume ventilation, early enteral nutrition, standardised antibiotic therapy for HAP, and systematic approach to extubation*
- In patients who develop severe acute respiratory distress syndrome (ARDS) and life-threatening hypoxaemia, rescue manoeuvres such as prone positioning and alveolar recruitment may be reasonable to improve oxygenation

Temperature

- Aim normothermia - target temperature 37.0C±0.5C
- Targeted Temperature Management (TTM) implemented with fever refractory to antipyretic medications during acute phase - *TTM should be initiated as rapidly as possible once fever detected if pharmacological treatment has not controlled temperature **within 1h** of administration - 1st line paracetamol / 1g / qds (see protocol)*

- Full **clerking** prior to angiography – *with full **neurological examination** if possible*
- Patient should be on **bed rest**
- Surgical team **book angiography**
- **Routine bloods i**ncluding **eGFR**
- **16G cannula** (grey) required
- Angiography +/- proceed to definitive treatment within 24h of admission to NHNN

Class of Recommendation

- **Class 1: STRONG** *Recommended (Benefit >>> Risk)*
- **Class 2a: MODERATE** *Reasonable (Benefit >> Risk)*
- **Class 2b: WEAK** *May be reasonable* (Benefit ≥ Risk)
- **Class 3: STRONG** *Potential harm (Risk > Benefit)*

Nimodipine 60mg / 4 hourly NG *or* PO

Early enteral nimodipine beneficial in preventing DCI and improving outcome. Consistent administration recommended even in cases of nimodipine-induced hypotension

- Commence as soon as SAH diagnosed
- Continue for 21 days
- If hypotensive after dose, change to 30mg / 2 hourly
- If not absorbing, give IV (1-2mg/h) via dedicated CVC lumen *(to run concurrently with 0.9% saline @ 40mL/h)*

Intravascular Volume and **Electrolyte Management**

- Aim for **EUVOLAEMIA** (2-3L/24h adequate for most)
- **Hypervolaemia** potentially harmful - *association with excess morbidity*
- Use of mineralocorticoids is reasonable to treat natriuresis and hyponatraemia *(see protocol)*

Glycaemic control

- Aim glucose 7-12mmol/L *(see protocol)*
- Strict hyperglycaemia management, and avoidance of hypoglycaemia may improve outcome

Venous thromboembolism (VTE) **Prophylaxis**

- Intermittent pneumatic compression device +/- TEDs
- Withhold LMWH if aneurysm unprotected
- 24h after aneurysm secured, pharmacological prophylaxis recommended to reduce risk for VTE
- LMWH should be withheld 24h before and 24h after any subsequent intracranial procedures

Analgesia / Anti-emetic

- NSAIDs not recommended for analgesia prior to treating aneurysm, but consider once aneurysm treated
- Cyclizine for nausea
- Avoid ondansetron due to susceptibility to QT interval prolongation

Seizure Management – Anti-epileptic Drugs (AEDs)

- **Routine** prophylactic AEDs not recommended
- Prophylactic AEDs may be reasonable if **high seizure risk features** - *Ruptured MCA aneurysm, high-grade SAH, ICH, hydrocephalus, cortical infarct*
- Treatment with AEDs for ≤7 days is reasonable in patients **presenting with seizures**
- Treatment with AEDs for >7days not beneficial
- Phenytoin **not** recommended for prophylaxis - *Due to associated cognitive effects and poor outcome*
- Other AEDs can be considered - *If so, short course of 3-7 days Levetiracetam (Keppra) recommended*
- Local practice should determine duration of AEDs *– but should be reviewed at 3 months*